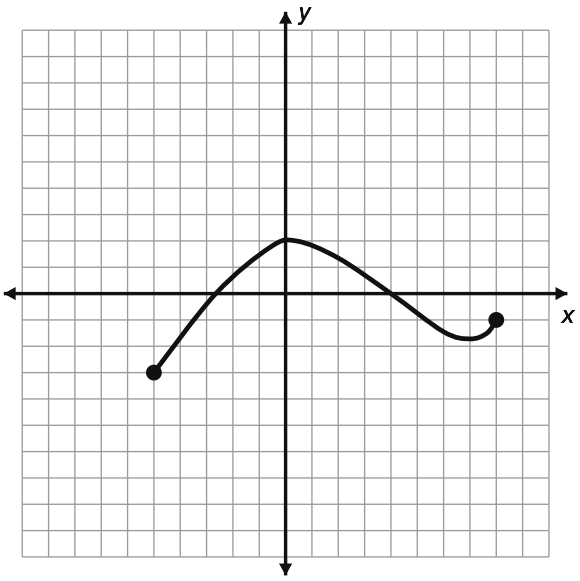
STATION

5

UNIT 5 REVIEW #2

Reading a graph

The graph below represents the function $y = f(x)$.



a) State the domain and range of this function.

A) Domain: $-5 \le x \le 8$
Range: $-3 \le y \le 2$

B) Domain: $-3 \le x \le 2$
Range: $-5 \le y \le 8$

C) Domain: $-5 \le x \le 8$
Range: $-3 \le y \le -1$

D) Domain: $-10 \le x \le 10$
Range: $-10 \le y \le 10$

b) What is the value of x when $f(x) = -3$?

c) What is $f(4)$?

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STATION

6

UNIT 5 REVIEW #2

Subtracting polynomials

What is $(x - 4)^2$ subtracted from $(x^2 + 3x - 4)$?

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STATION

9

UNIT 5 REVIEW #2

Terms from a formula

Find the first 5 terms given the formula $f(n) = 3(n - 1) + 4$

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STATION

10

UNIT 5 REVIEW #2

Terms from a formula

Find the first 5 terms given the formula $f(n) = 2(5)^{n-1}$

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STATION

11

UNIT 5 REVIEW #2

Sequence: 1, 2, 4, 8, ...

Given the following: 1, 2, 4, 8, 16, 32, ...

a) Is the pattern Arithmetic, Geometric, or Neither?

A) Arithmetic

B) Geometric

C) Neither

b) Write a rule that represents the *n*th term of the sequence.

c) Find the 10th term of the sequence.

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STATION

12

UNIT 5 REVIEW #2

Sequence: 4, 7, 10, ...

Given the following: 4, 7, 10, 13, 16, ...

a) Is the pattern Arithmetic, Geometric, or Neither?

A) Arithmetic

B) Geometric

C) Neither

b) Write a rule that represents the *n*th term of the sequence.

c) Find the 12th term of the sequence.

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STATION 13

UNIT 5 REVIEW #2

Build an arithmetic sequence

Write an arithmetic sequence in which $a_1 = 3$ and $a_4 = 15$. Find the 13th term.

a) Write the sequence.

b) Find the 13th term.

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STATION

14

UNIT 5 REVIEW #2

Geometric sequence with fractions

The terms of a geometric sequence are given below.

$1,\ \frac{1}{3},\ \frac{1}{9},\ \frac{1}{27}, \ldots$

a) Write a rule for the nth term of this geometric sequence.

b) Determine the 7th term.

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STATION

17

UNIT 5 REVIEW #2

Common ratio of −2

A geometric sequence with a common ratio of −2 is

a) 6, 4, 2, 0, ...

b) 5, 7, 9, 11, ...

c) −3, 6, −12, 24, ...

d) −10, −8, −6, −4, ...

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STATION

18

UNIT 5 REVIEW #2

Solve for x

If $f(x) = \frac{5x - 9}{2}$, algebraically determine the value of x when $f(x) = 8$.

Show every step.

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